

Power-law relations

Log-log regressions

- Several biological relations take the form of power-law relations
- We can linearize these relations using a log-log transformation

$$y \propto ax^b$$



Take the log on both sides

$$\log(y) \propto \log(ax^b) = \log(a) + \log(x^b) =$$

$$\log(y) \propto \log(a) + b \log(x)$$

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Log-log regressions

- Several biological relations take the form of power-law relations
- We can linearize these relations using a log-log transformation
- In this model, the slope is an estimate of the exponent of the power-law
- The interpretation of the slope is that a 1% increase in x leads to a β % increase in y

$$\log(y_i) \sim N(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta \log(x_i)$$



$$\% \Delta y \approx \beta \% \Delta x$$